

TIER 1 PROJECT

Pixel Predators

Plant Disease Classification

ITAI 1378 | Midterm Project Proposal

TEAM MEMBERS

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Upload a leaf photo. Get a disease label and confidence score.

Image: Septoria leaf spot of tomato, Wikimedia Commons/Bugwood.org, CC BY 3.0.



The Problem



>\$220B

annual global losses from plant
pests and diseases



Up to 40%

of crop production lost to pests and
diseases each year



Manual

scouting can be slow, costly, and
dependent on expert access

Who cares: Farmers, agronomists, food security groups, and crop insurance teams.

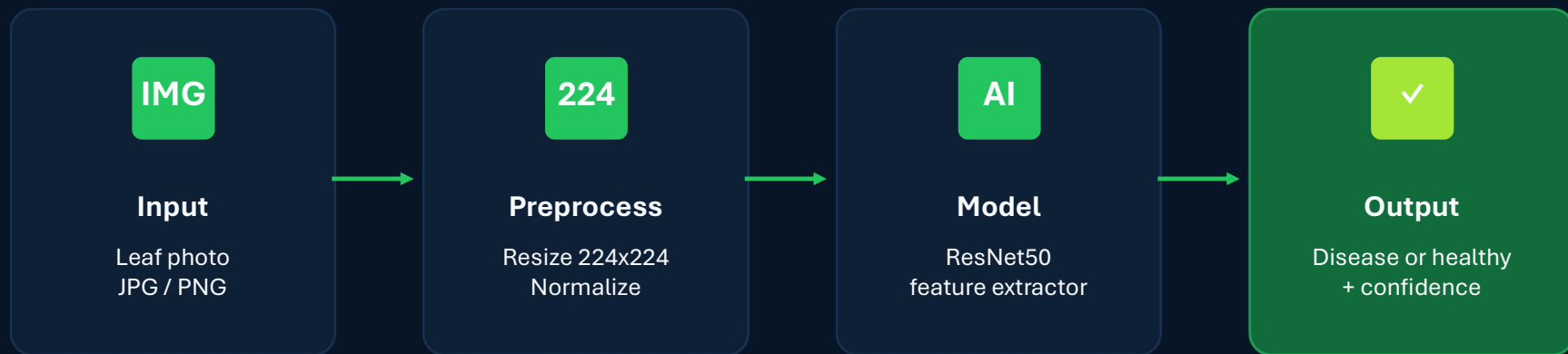
What it solves: Fast preliminary screening from a leaf photo, helping growers make earlier treatment decisions.



Sources: FAO Plant Production and Protection facts and figures; USDA ARS notes that early detection and rapid response are key to mitigating plant diseases.

Your Solution

A transfer-learning app that classifies a leaf image and returns the predicted plant condition with a confidence score.



Application layer:

Gradio upload page - user selects a leaf image - app shows top prediction, confidence, and a simple report.

Model: ResNet50 pretrained on ImageNet | Framework: PyTorch + torchvision | Tier: 1



Technical Approach

AI

CV Technique

Image classification using transfer learning

M

Model

ResNet50, pretrained on ImageNet; freeze most backbone layers and fine-tune the final classifier

FW

Framework

PyTorch + torchvision for training; Hugging Face or Kaggle for dataset access; Gradio for demo UI

✓

Why this fits

Classification matches the goal: one image in, one crop-condition label out. ResNet50 is strong, simple, and realistic on free Colab.



Data Plan

DATASET

PlantVillage Dataset

Public dataset via Kaggle / TensorFlow Datasets / original PlantVillage source

- ~54K healthy and diseased leaf images
- 38 crop-condition classes across 14 crop species
- Labels include crop name + disease/healthy status
- Mostly controlled leaf photos, so real-world testing is important
- Fallback: balanced 10-class subset if Colab limits hit

PREPROCESSING + SPLIT

01

Resize

224 x 224 px for ResNet50 input

02

Augment

Training only: flip, rotation, color jitter

03

Normalize

ImageNet mean and standard deviation

04

Split

70% train / 15% validation / 15% test



Success Metrics



PRIMARY METRIC

90%+

Top-1 test accuracy

F1

Macro F1

≥ 0.88 average across classes



Speed

< 1 second per image

R

Recall check

Review low-performing classes

CM

Tools

Confusion matrix + error samples

Final report will use the held-out test set. Validation will be used only for model tuning and parameter choices.

Milestone Plan



6-week team plan from project setup to final presentation and submission.

Week 1	Create Team	Form the Pixel Predators team and confirm team members
Week 2	Blueprint	Review requirements, start the proposal, GitHub repo, and README
Week 3	Choose Project + Dataset	Select plant disease classification and confirm the PlantVillage dataset
Week 4	Refine Project + Assign Tasks	Finalize scope, assign team roles/titles, and divide slide/project tasks
Week 5	Improve Project	Improve the model plan, metrics, risks, resources, and demo workflow
Week 6	Present and Submit	Finalize slides, repository, proposal PDF, presentation, and Canvas submission

This plan uses six project weeks; exact calendar dates should match Canvas.

Risks and Resources



Top Risk 1

PlantVillage images may not match outdoor field photos.

Plan B: use stronger augmentation, test a few real leaf photos, and clearly state the app is a preliminary screening tool.

Top Risk 2

Full dataset may exceed free Colab time or memory.

Plan B: train a balanced 10-class subset, lower batch size, save checkpoints to Drive, and use Kaggle Notebooks as backup.

Resources Needed

- Compute: Google Colab free tier; Kaggle Notebooks backup
- Storage: Google Drive for model checkpoints
- Tools: PyTorch, torchvision, Hugging Face Datasets, Gradio
- Data: PlantVillage public dataset
- Estimated cost: \$0

Fully achievable within the term using free tools.